

Financial analytics of BTC inverse options in a stochastic volatility world

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Abstract

Bitcoin (BTC) has attracted a plethora of investors and professional traders and has become an almost inevitable asset class in today's financial markets. Deribit, a centralized exchange, dominates about 90% of the global crypto options market, where, notably, BTC options are settled in BTC. Analytical explorations into BTC option pricing are limited. In this paper, we present the Stochastic Volatility with Correlated Jumps (SVCJ) model, enhancing BTC options' pricing. We refine simulation methods to calibrate the SVCJ for BTC options, providing a Delta hedging formula. We find that the SVCJ model shows the best fitting properties in in-sample and out-of-sample performance over simpler models. It also maintains consistent hedging accuracy dynamically. Our findings offer substantial insights for effective BTC options trading, paving the way for improved risk management and investment strategies.

Key words: Bitcoin, crypto options, stochastic volatility models, correlated jumps, Delta hedge

Introduction

Cryptocurrency (CC) is a digital currency based on blockchain technology, which offers a unique way of securely recording and transferring information, such as a ledger of payments (Pagnottoni and Dimpfl, 2019). Since the advent of Bitcoin (BTC) as the first cryptocurrency by Nakamoto (2008), the rapidly expanding CC market has enticed numerous investors. With the collective market capitalization of various CCs now comparable to the top ten global stocks, BTC has emerged as a pivotal asset, necessitating analytical insights into its derivatives market. Consequently, this is the focus of this research: How to price and hedge BTC options?

Option pricing has been a highly active research field since the seminal work by Black and Scholes Black and Scholes (1973). However, despite the increasing popularity of cryptocurrency derivatives, a widely accepted valuation model is still

lacking (Cao and Celik, 2021). To address this gap, Cao (2001) extended existing frameworks and provided formulas for pricing European options on BTC within a small open economy, treating BTC as the equilibrium exchange rate. Furthermore, Alexander et al. (2023) emphasized that pricing BTC options should be approached either from the perspective of pricing foreign exchange (FX) options, as suggested by Garman and Kohlhagen (1983), or within an incomplete market setting, considering that the underlying Deribit BTC index is non-tradable.

A comprehensive understanding of BTC dynamics is essential for pricing options and other financial derivatives linked to this cryptocurrency. Researchers such as Bariviera et al. (2017) have examined and compared long-term memory and other statistical properties of BTC dynamics with those of standard currencies. Their findings shed light on the unique characteristics of BTC and its potential implications for option pricing. Additionally, Scaillet et al. (2020) analyze tick-level price dynamics in BTC, discovering that jumps occur more frequently in BTC compared to US large-cap stocks and equity indices, further highlighting the distinct features of BTC dynamics.

The existing literature demonstrates the necessity of studying BTC dynamics to price BTC options and other related financial instruments effectively. However, the lack of a universally accepted valuation model and the unique properties of BTC pose challenges that require further exploration and analysis.

The first approach in analyzing CC options estimates model parameters using historical BTC returns and calculates model option prices based on these parameters. For example, Hou et al. (2020) estimate stochastic volatility with correlated jumps models with the physical measure using historical BTC returns, then simulate option prices for further sensitivity analysis. Cretarola et al. (2020) propose a bivariate model to include market attention as an exogenous factor as an extension to the geometric Brownian motion and provide a closed pricing formula for European-style Bitcoin options using the Profile Quasi Maximum Likelihood method. Empirical comparisons using USD-denominated BTC options from Deribit. Jalan et al. (2021) estimate the BS model and Heston-Nandi GARCH model with regimes switching in volatility using historical BTC returns and compare model prices using these estimates with USD-denominated BTC options from Deribit. It is found that the Heston-Nandi model predicts prices more accurately than the BS model. Siu and Elliot (2021) estimate a nonlinear time series model combining both the self-exciting threshold autoregressive (SETAR) model and the generalized autoregressive conditional heteroscedastic (GARCH) model using BTC return dynamics. Then, the conditional Esscher transform and the variance-dependent pricing kernel are used to generate BTC Options for sensitivity analysis. Chen and Huang (2021) find that including the jumps model in returns and volatilities are significant in BTC prices through Bitcoin Price Index (BPI) traded daily on CoinDesk, and derive a quasi-closed formula for vanilla Bitcoin options and their Greeks based on risk-neutral measures.

An alternative approach is calibrating risk-neutral model parameters using options data by minimizing the difference between model and market option prices (Bakshi et al., 1997). This approach usually needs a closed-form pricing formula and challenges numerical stability in model calibration. Therefore, this approach is currently limited

to vanilla options and simple models. For example, under the Black-Scholes (BS) model, Zulfikar and Gulzar (2021) study patterns of implied volatility for Deribit USD-denominated options from Deribit based on short-dated options and show that BTC options belong to the commodity class of assets because of the presence of volatility forward skew. Madan et al. (2019) calibrate a series of Markov models: the Black-Scholes model, Laplace model, five variance gamma-related models, and the Heston model, through BTC vanilla options available on LedgerX, and examine their pricing performance.

Following this research streamline, the novelty of this paper is four-folded. First, we focus on BTC-denominated options in Deribit for its dominance in open interests and because its settlement involves the Deribit BTC index. Second, we focus on the stochastic volatility and jumps model because it explains the BTC dynamics well. Third, we provide a feasible simulation scheme to calibrate the more complex SVCJ model using Deribit BTC-denominated options under the risk-neutral probability measure: This allows us to compare whether the more complex SVCJ model provides better fits to market data, investigate the dynamic of implied parameters and whether the more complex SVCJ model helps in pricing and hedging. Fourth, with the calibrated parameters, we provide the Delta formula to implement dynamic hedging routines (Hull, 2014).

The rest of this paper is organized as follows. Section 2 provides an overview of BTC options based on data from the Blockchain-Research-Center.com. Section 3 provides a feasible simulation scheme to calibrate option pricing models using Deribit BTC-denominated options. This allows us to investigate implied parameters of the stochastic volatility with the correlated jumps model and its nested models. In-sample and out-of-sample pricing errors among these models are compared. Section 4 provides the Delta formula and implements dynamic Delta hedging. The last section concludes. All code for this paper is available on quantlet.com, and a courselet video is available on quantinar.com.

Deribit BTC options

The centralized Deribit option exchange, currently dominating the crypto options market with approximately 90% of the open interest, outshines competitors like CME, accounting for only around 8% of the total market share¹. Within Deribit, BTC options are quoted in both BTC and USD. Upon the contract's exercise, settlement occurs immediately after expiry, based on the Deribit BTC index. The index is computed as a weighted average of adjusted prices from Bitstamp, Coinbase Pro, Kraken, and LMAX Digital. The exercise-settlement value (S_T) represents the average Deribit index value during the final 30 minutes preceding expiry. Define K as the strike price and T as the time to maturity, expressed in days. The settlement amount for a BTC-denominated option is calculated by dividing the difference by the exercise value and exhibits the following payoff:

$$\frac{1}{S_T} \max \{ \omega(S_T - K), 0 \}, \omega = \pm 1, \quad (1)$$

¹ Source: <https://coinglass.com/options>

where we use $\omega = 1$ to denote a call option and $\omega = -1$ for a put option. We refer to BTC-denominated options as "inverse options" because (1) is associated with the inverse of the BTC index.

When BTC options are quoted in USD on Deribit, their calculation involves the use of the exercise-settlement value and the mark price, which serves as an exchange rate between BTC and USD. The mark price represents the valuation of the futures contract during trading hours and is determined based on the last traded futures price unless it falls outside the range defined by the current best bid and ask prices. The mark price is computed using the following formula:

Mark Price = Index Price + 30-second EMA of (Futures Mark Prices - Index Price),

where EMA refers to the exponential moving average. Let F_T denote the mark price at maturity. As F_T differs from S_T , the USD-denominated payoff for BTC options becomes:

$$F_T \left\{ \frac{1}{S_T} \max \{ \omega(S_T - K), 0 \} \right\}, \omega = \pm 1. \quad (2)$$

Pricing USD-denominated options involves the consideration of bivariate dynamics between F_T and S_T , making the process more complex. For simplicity, our focus remains on BTC-denominated options with the payoff defined in Equation (1). Additional information on options contract specifications can be found in the Deribit knowledge base, accessible at <https://www.deribit.com/kb/options>.

We conduct now a comprehensive analysis of BTC options. Figure 1 depicts the BTC-denominated payoff function for call and put options, showcasing the relationship with S_T . The visual representation emphasizes the distinct characteristics of BTC call and put options. Notably, the call option exhibits a capped payoff at one BTC, whereas the put option demonstrates the unlimited potential for a positive payoff. This graphical depiction effectively highlights the unique features of these two option types within the Bitcoin-denominated context.

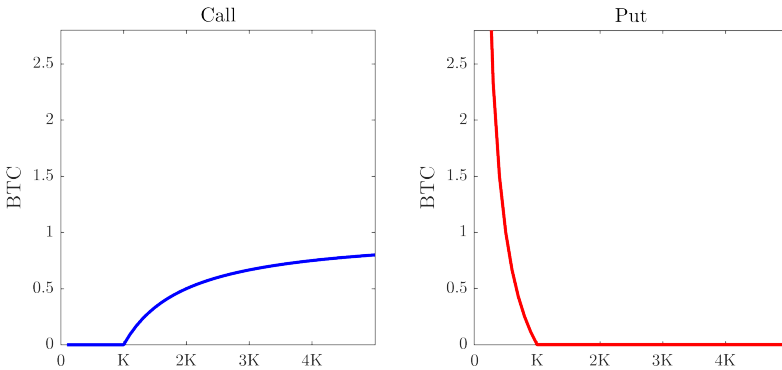


Fig. 1: Payoff functions of inverse options. This figure compares the payoff settled in BTC against S_T for both BTC call and put options, given a strike price of K USD.

 Deribit_inverse_BTC_options

As data service we use nice transaction data for inverse options, obtained from Blockchain-Research-Center.com, covering the period from December 1, 2021, to February 28, 2022. The dataset consists of 736,563 observations across 90 trading days. As of September 2, 2022, the total open interest for BTC options stands at 243.60K BTC, with Deribit commanding a dominant market share of 88% (215.90K BTC), followed by CME at 8% (16.44K BTC) and Okex at 3% (8.44K BTC).

Table 5 in the Appendix summarizes the columns in the dataset, providing explanations for each. The intraday transaction data includes timestamps, BTC prices in USD, implied volatility, option types, strike prices, and maturities for inverse options. This comprehensive dataset enables a thorough examination of the dynamics within the BTC options market.

Figure 2 illustrates the time series of daily cumulative numbers of trades and cumulative daily volume for inverse options based on the intraday transaction data. The graph depicts the daily number of trades and volume stability throughout the specified period. On average, there are 8,184 trades per day, with a daily volume of 21,155.

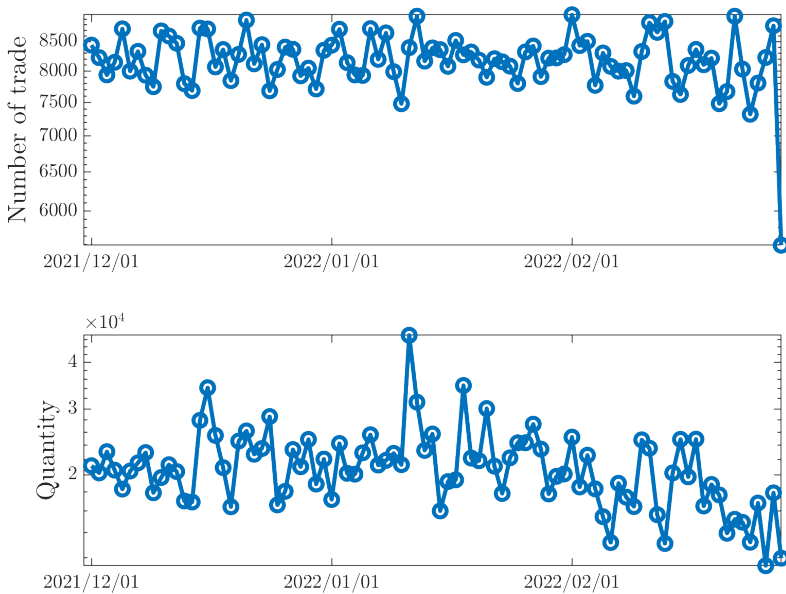


Fig. 2: **Daily accumulated trades and quantities.** For intraday transactions on inverse options, the upper panel provides a time series plot for the daily accumulative number of trades, and the lower panel provides a time series plot for daily accumulated quantities.  **Deribit.inverse.BTC_options**

Figure 3 illustrates the strike prices of intraday transactions for BTC call and put options during the study period and superimposes the intraday BTC prices as a bold

black line. Two key observations can be made: (1) Out-of-the-money (OTM) options are traded more frequently than in-the-money (ITM) options. Specifically, inverse BTC call options tend to be traded more often at strike prices above current BTC prices, while inverse BTC put options are more frequently traded at strike prices below current BTC prices. (2) The range of strike prices is considerably broader than that observed in traditional options markets. The average BTC price is 43,874.72 USD, yet the minimum and maximum strike prices for inverse options are 12,000 USD and 400,000 USD, respectively.

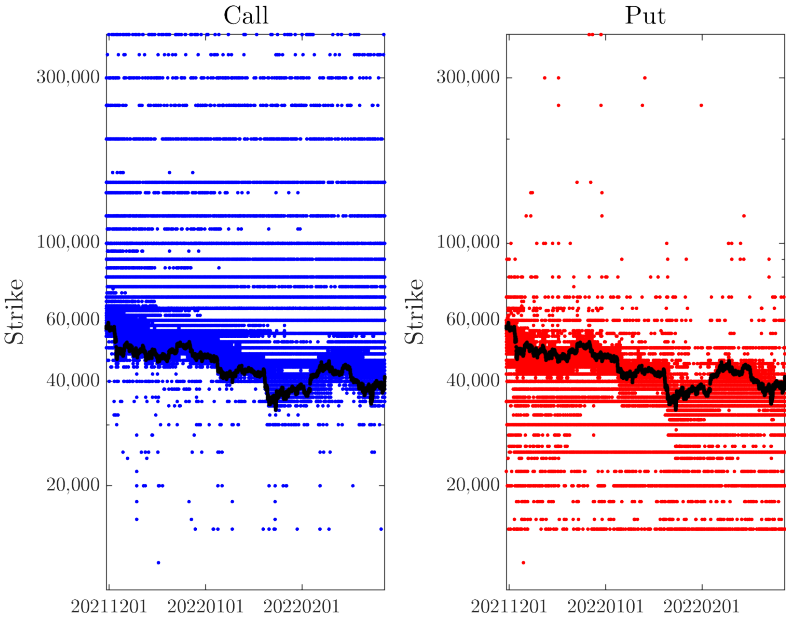


Fig. 3: **Visualizing strike prices for the intraday transaction of inverse options.** The left and right panels display strike prices for intraday transactions of BTC call and put options. A bold black line represents the BTC prices.  Deribit_inverse_BTC_options

Figure 4 presents a three-dimensional scatter plot illustrating the prices of inverse options traded on January 18, 2022. The plot reveals that inverse options are predominantly traded with short-term maturity across various moneyness levels. Moreover, excluding the top 0.1% of BTC inverse option prices, all remaining options have prices below 0.2 BTC.

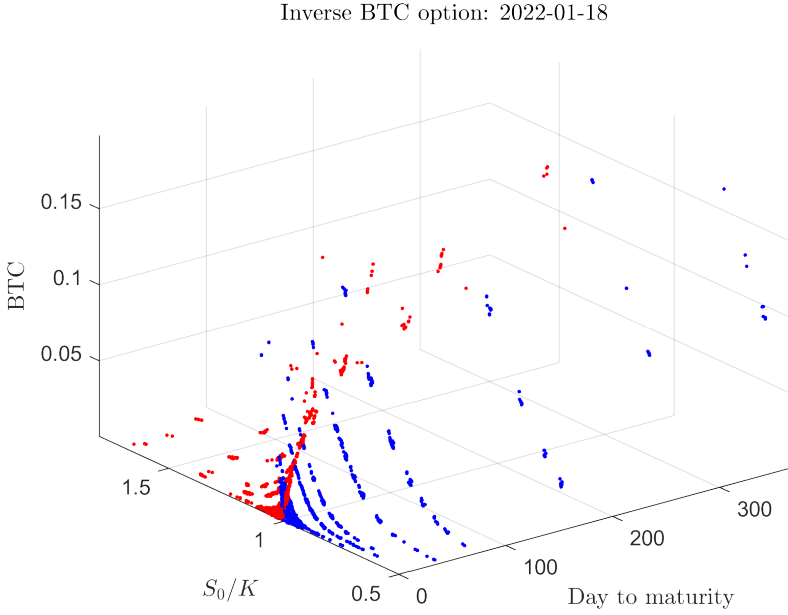


Fig. 4: **Visualizing prices of inverse options.** This figure presents a 3-D scatter plot illustrating the prices of inverse options traded on January 18, 2022. Call options are depicted in blue, while put options are shown in red. We note that the highest 0.1% of BTC inverse option prices have been removed from the display for better visualization.  Deribit_inverse.BTC_options

Table 1 presents percentiles of time to maturity (T), moneyness (K/S_0), and prices for intraday transactions. Approximately 50% of the options are short-term: the median time to maturity is nine days for inverse call options and seven days for inverse put options. Furthermore, over 90% of the options are traded out-of-the-money (OTM). However, 5% of the options are traded at extremely high or low strike prices: the 5th percentile of moneyness for call options is 0.51, while the 95th percentile for put options is 1.55.

Table 1. Distributions of option characteristics. The table presents percentiles for BTC call and put options' time to maturity (T), moneyness (S_0/K), and prices (p), with Q_k representing the k -th percentile. The analysis covers the period from December 1, 2021, to February 28, 2022.

 Deribit_inverse.BTC_options

Percentile	T		S_0/K		p	
	Call	Put	Call	Put	Call	Put
Min	0	0	0.09	0.12	0.0001	0.0002
Q_1	0	0	0.27	0.85	0.0005	0.0005
Q_5	1	0	0.51	0.96	0.0005	0.0010
Q_{10}	1	1	0.64	0.99	0.0010	0.0015
Q_{20}	1	1	0.78	1.00	0.0030	0.0035
Q_{30}	2	2	0.86	1.02	0.0055	0.0055
Q_{40}	6	4	0.91	1.03	0.0080	0.0090
Q_{50}	9	7	0.94	1.05	0.0115	0.0130
Q_{60}	15	12	0.96	1.08	0.0170	0.0190
Q_{70}	22	18	0.98	1.12	0.0250	0.0280
Q_{80}	42	30	0.99	1.21	0.0380	0.0435
Q_{90}	84	64	1.00	1.38	0.0630	0.0740
Q_{95}	149	105	1.02	1.55	0.0954	0.1095
Q_{99}	291	259	1.06	2.11	0.1750	0.2275
$Q_{99.9}$	355	347	1.34	2.91	0.3260	0.6375
$Q_{99.99}$	363	364	2.63	3.38	0.6255	4.2101
Max	365	365	3.90	4.04	0.7465	7.3356

As previously mentioned, the options traded for BTC have a broader range of moneyness compared to standard options. Therefore, we establish maturity-moneyness categories as shown in Table 2 and report the percentage of traded options in each category. Call options account for 55.6% of the trades, and out-of-the-money (OTM) options make up 82.3%. The following section provides an overview of the main empirical results.

Table 2. Maturity-moneyness categories. The table delineates nine categories based on maturity-moneyness. Specifically, short-term, mid-term, and long-term options are defined by their time to maturity (T), while OTM, ITM, and deep ITM options are defined by their moneyness (S_0/K). Additionally, the table includes the proportion (%) of trades for inverse options in each maturity-moneyness category.  Deribit_inverse_BTC_options

Inverse BTC call option	Deep OTM	OTM	ITM	Sum
	$\frac{S_0}{K} \leq 0.86$	$0.86 < \frac{S_0}{K} \leq 1$	$1 < \frac{S_0}{K}$	
Short-term: $T \leq 5$	0.3%	16.3%	3.7%	20.3%
Mid-term: $5 < T \leq 28$	6.0%	12.9%	1.8%	20.7%
Long-term: $28 < T$	10.9%	3.0%	0.7%	14.6%
Sum	17.2%	32.2%	6.2%	55.6%
Inverse BTC put option	ITM	OTM	Deep OTM	Sum
	$\frac{S_0}{K} \leq 1$	$1 < \frac{S_0}{K} \leq 1.12$	$1.12 < \frac{S_0}{K}$	
Short-term: $T \leq 5$	3.5%	13.5%	1.1%	18.1%
Mid-term: $5 < T \leq 28$	2.5%	7.8%	6.4%	16.7%
Long-term: $28 < T$	1.3%	2.3%	6.0%	9.6%
Sum	7.3%	23.6%	13.5%	44.4%

Table 3 presents the mean prices and standard deviations (in parenthesis) for call and put options across each maturity-moneyness category. For call options, the average prices are less than one BTC across all maturity-moneyness categories. Additionally, it aligns with expectations that BTC options are pricier for longer-term and deeper ITM positions, both for call and put options.

Table 3. Average prices in BTC and standard deviations of inverse options. This table displays the mean prices and standard deviations (in parenthesis) for options across each maturity-moneyness category.  Deribit_inverse_BTC_options

Inverse BTC call option	Deep OTM	OTM	ITM
	$\frac{S_0}{K} \leq 0.86$	$0.86 < \frac{S_0}{K} \leq 1$	$1 < \frac{S_0}{K}$
Short-term: $T \leq 5$	0.0006	0.0054	0.0232
	(0.0003)	(0.0049)	(0.0190)
Mid-term: $5 < T \leq 28$	0.0059	0.0224	0.0683
	(0.0059)	(0.0154)	(0.0468)
Long-term: $28 < T$	0.0358	0.0865	0.1614
	(0.0391)	(0.0477)	(0.0946)
Inverse BTC put option	ITM	OTM	Deep OTM
	$\frac{S_0}{K} \leq 1$	$1 < \frac{S_0}{K} \leq 1.12$	$1.12 < \frac{S_0}{K}$
Short-term: $T \leq 5$	0.0172	0.0041	0.0013
	(0.0558)	(0.0052)	(0.0011)
Mid-term: $5 < T \leq 28$	0.0695	0.0234	0.0071
	(0.0815)	(0.0145)	(0.0062)
Long-term: $28 < T$	0.1884	0.0816	0.0342
	(0.3674)	(0.0373)	(0.0289)

It is intriguing to observe that a put option can have a value of more than one BTC. Only a tiny fraction, about 0.04% (130 put options), have prices exceeding one BTC. Upon examining the distribution of these exceedingly high-priced put options, we found that they are mostly long-term and deep in the money (with moneyness between 0.5 to 0.1). This data implies that investors anticipate a possibility of the BTC price decreasing to half or one-tenth of its present value, respectively.

Pricing BTC-denominated options

This section is structured as follows: Section 3.1 introduces the stochastic volatility with the correlated jumps model and its nested variants. Section 3.2 presents a practical simulation scheme for calibrating the model. In-sample and out-of-sample pricing errors among different models are compared in Section 3.3, while implied parameters are investigated in Section 3.4.

The stochastic volatility with correlated jumps model

In this section, we focus on pricing and analyzing the stochastic volatility with correlated jumps (SVCJ) model, as recommended by Hou et al. (2020) for suitable stochastic dynamics. Denote r as the discount rate. The SVCJ model is formulated as follows:

$$\frac{dS_t}{S_t} = rdt + \sqrt{V_t}dW_t^s + Z_t^y dN_t \quad (3)$$

$$dV_t = \kappa(\theta - V_t)dt + \sigma_v \sqrt{V_t}dW_t^v + Z_t^v dN_t \quad (4)$$

$$\text{Cov}(dW_t^s, dW_t^v) = \rho dt \quad (5)$$

$$P(dN_t = 1) = \lambda dt \quad (6)$$

$$Z_t^y | Z_t^v \sim N(\mu_y + \rho_j Z_t^v, \sigma_y^2), \quad (7)$$

$$Z_t^v \sim \text{Exp}(\mu_v). \quad (8)$$

Considering our focus on short-term options in a nearly risk-free environment, we set $r = 0$ following the approach of Matic et al. (2023).

In this SVCJ model, the parameters can be understood as follows. The correlation between the two Brownian motions, W_t^s and W_t^v , in the return process (3) and variance process (4) is represented by ρ . In the variance process, κ denotes the speed of mean reversion towards the long-term variance, θ signifies the long-term variance, and σ_v represents the volatility of the variance process. The jump components in the return and variance processes exhibit a jump frequency of λ , as indicated in (6). The jump size Z_t^v in the variance process must be positive and is assumed to follow an exponential distribution with a positive mean of μ_v . Given the jump size of the variance process, Z_t^y , the jump size in the return process, Z_t^y , is assumed to have a normal distribution with a mean of $\mu_y + \rho_j Z_t^v$ and a variance of σ_y^2 .

This SVCJ model encompasses a diverse array of nested models. Various stochastic volatility or jump-diffusion models can be derived by imposing certain constraints on the SVCJ model's parameters. For instance, if we set Z_t^v in (8) to zero, jumps only occur in prices, and the model simplifies to the SVJ model proposed by Bates (1996). If we set $\lambda = 0$, the model lacks jumps and reverts to the original SV model introduced by Heston (1993). By setting $\kappa = \theta = \sigma_v = 0$ and defining $Z_t^v = 0$, the model reduces to the pure jump-diffusion model presented by Merton (1976). Furthermore, the pure jump-diffusion model of Merton (1976) simplifies to the renowned BS model (Black and Scholes, 1973) when $Z_t^y = 0$.

Let $\Theta = (\theta_1, \dots, \theta_p)$ represent the parameter vector of size p . Since stochastic volatility is a prominent feature in financial time series data, we concentrate on the SV, SVJ, and SVCJ models. Specifically, for the SV model, we have $\Theta = (\mu, \rho, \alpha, \beta, V_0, \sigma_v)$ with $p = 6$. For the SVJ model, $\Theta = (\mu, \rho, \alpha, \beta, V_0, \sigma_v, \lambda, \mu_y, \sigma_y)$ with $p = 9$. Lastly, for the SVCJ model, $\Theta = (\mu, \rho, \alpha, \beta, V_0, \sigma_v, \lambda, \mu_y, \rho_j, \sigma_y, \mu_v)$ with $p = 11$.

Model calibration

Let $p(\omega_i, T_i, K_i)$ and $m(\Theta; \omega_i, T_i, K_i)$ represent the i -th observed market and model option price, respectively. Define the loss function as the L_2 distance between market and model prices (Bakshi et al., 1997),

$$\mathcal{L}(\Theta) = \sum_i^M \{p(\omega_i, T_i, K_i) - m(\Theta; \omega_i, T_i, K_i)\}^2, \quad (9)$$

where M denotes the number of observed options. The parameter Θ^* is calibrated as the solution to the constrained optimization problem:

$$\Theta^* = \arg \min_{\Theta \in \mathcal{A}} \mathcal{L}(\Theta), \quad (10)$$

where the set $\mathcal{A}$ for Θ is defined to guarantee the positiveness of the stochastic volatility models. We enforce constraints on the parameters rather than relying on the Feller condition $2\kappa\theta > \sigma_v^2$ (or $2\alpha > \sigma_v^2$ upon re-parametrization). Specifically, we require $\alpha > 0$, $\beta < 1$, and $V_0 > 0$. We also impose a natural constraint on the correlation coefficient between two Brownian motions, $-1 \leq \rho \leq 1$. For the SVJ and SVCJ models, λ represents the probability of a jump, and thus it must satisfy $0 \leq \lambda \leq 1$. In the case of the SVCJ model, the average jump size for volatility should meet the condition $\mu_v \geq 0$.

For stochastic volatility models, closed-form formulae for pricing inverse options are unavailable. As a result, Monte Carlo simulation must be employed to achieve this. We utilize the Euler-Maruyama discretization (??, bel). Define $Y_{t+1} = \log(S_{t+1}/S_t)$ as the log-return for $t = 1, \dots, T$. The SVCJ price path is determined through:

$$S_t = S_0 \exp(Y_1 + \dots + Y_t) \quad (11)$$

for $t = 1, \dots, T$, where S_0 represents the initial stock price. Let $\alpha = \kappa\theta$ and $\beta = 1 - \kappa$. Here, Y_t and V_t are generated as follows,

$$Y_t = r + \sqrt{V_{t-1}}\varepsilon_t^y + Z_t^y J_t, \quad (12)$$

$$V_t = \alpha + \beta V_{t-1} + \sigma_v \sqrt{V_{t-1}}\varepsilon_t^v + Z_t^v J_t, \quad (13)$$

where ε_t^y and ε_t^v are $N(0, 1)$ variables with correlation ρ , J_t is a Bernoulli random variable with $P(J_t = 1) = \lambda$, and jump sizes follow $Z_t^y | Z_t^v \sim N(\mu_y + \rho_j Z_t^v, \sigma_y^2)$ and $Z_t^v \sim \text{Exp}(\mu_v)$.

To overcome numerical instability caused by Monte Carlo simulation error in the approximated lass, denoted as $\hat{\mathcal{L}}(\Theta)$, we employ common random numbers for variance reduction (Glasserman, 2004). The strategy is to generate one set of random samples only one time and use it throughout the entire optimization process when approximating $m(\Theta; \omega_i, T_i, K_i)$. We utilize the inverse method for Bernoulli and exponential distributions (Ross, 2022), employing the quantile functions for the Bernoulli random variable with parameter λ , given by $I_{\{u > (1-\lambda)\}}(u)$, and for the exponential random variable with mean μ , given by $-\frac{\log(u)}{\mu_v}$. Additionally, we generate correlated normal random variables $\varepsilon_t^{v,(n)}$ and $\varepsilon_t^{y,(n)}$ from two independent normal random variables using Cholesky decomposition. Let N denote the simulation sample size.

- Step 1: Generate $U_{i,t}^{(n)} \stackrel{i.i.d.}{\sim} U(0, 1)$ for $i = 1, 2$ and $Z_{j,t}^{(n)} \stackrel{i.i.d.}{\sim} N(0, 1)$ for $j = 1, 2, 3$ for $t = 1, \dots, T$ and $n = 1, \dots, N$.
- Step 2: Calibrate Θ by solving the constrained optimization in (10), where $m(\Theta; \omega_i, T_i, K_i)$ is approximated by $\hat{m}(\Theta; \omega_i, T_i, K_i)$ using common random numbers generated from Step 1:

- a. Set $J_t^{(n)} = I_{\{U_{1,t}^{(n)} > (1-\lambda)\}} \left(U_{1,t}^{(n)} \right)$,
- b. Set $Z_t^{v,(n)} = -\frac{\log(U_{2,t}^{(n)})}{\mu_v}$,
- c. Set $Z_t^{y,(n)} = (\mu_y + \rho_j Z_t^{v,(n)}) + \sigma_y Z_{1,t}^{(n)}$,
- d. Set $\varepsilon_t^{v,(n)} = Z_{2,t}^{(n)}$,
- e. Set $\varepsilon_t^{y,(n)} = \rho Z_{2,t}^{(n)} + \sqrt{1-\rho^2} Z_{3,t}^{(n)}$,
- f. Calculate $(Y_t^{(n)}, V_t^{(n)})$ using (12) and (13) for $t = 1, \dots, T$ and $n = 1, \dots, N$.
- g. Calculate $S_t^{(n)}$ using (11) for $t = 1, \dots, T$ and $n = 1, \dots, N$.
- h. Compute the BTC option price with the sample mean,

$$\hat{m}(\Theta; \omega, T, K) = \frac{e^{-rT}}{N} \sum_{n=1}^N \frac{1}{S_T^{(n)}} \max \left\{ \omega(S_T^{(n)} - K), 0 \right\}.$$

Using common random numbers is a straightforward and effective method for calibrating implied parameters in options pricing. A related use of common random numbers for identifying optimal tilting parameters in importance sampling is discussed in Teng (2023). This connection highlights the versatility of common random numbers to address various computational challenges in financial model calibration and other related fields.

Our research methodology is outlined as follows. We consider all intraday transactions of inverse options and calibrate a distinct set of model parameters for each day. It is worth noting that all analyses throughout this paper are conducted using Matlab (R2021a), and we employ a Monte Carlo sample size of $N = 20,000$. For the calibration of the BS model, we utilize the Matlab solver `fminsearch`. In the case of the SV, SVJ, and SVCJ models, we use the Matlab solver `lsqnonlin` for optimization with the Levenberg-Marquardt algorithm, setting the 'FunctionTolerance' to 10^{-4} .

In-sample and Out-of-sample pricing errors

To assess the goodness of fit for a model applied to BTC options, we compute the in-sample root-mean-squared-error (RMSE). The in-sample RMSE is given by

$$\text{In-sample RMSE} = \sqrt{\frac{1}{M} \sum_{i=1}^M \{p(\omega_i, T_i, K_i) - \hat{m}(\Theta^*; \omega_i, T_i, K_i)\}^2}.$$

A model yielding a smaller in-sample RMSE is preferable as it generates option prices closer to the observed market option prices.

Figure 5 presents boxplots and time series plots of the in-sample RMSE for the BS, SV, SVJ, and SVCJ models. The time series plots of the in-sample RMSE across these four models exhibit a notable consistency: the SVCJ model outperforms the others by producing the lowest in-sample RMSE, followed by the SVJ, SV, and BS models.

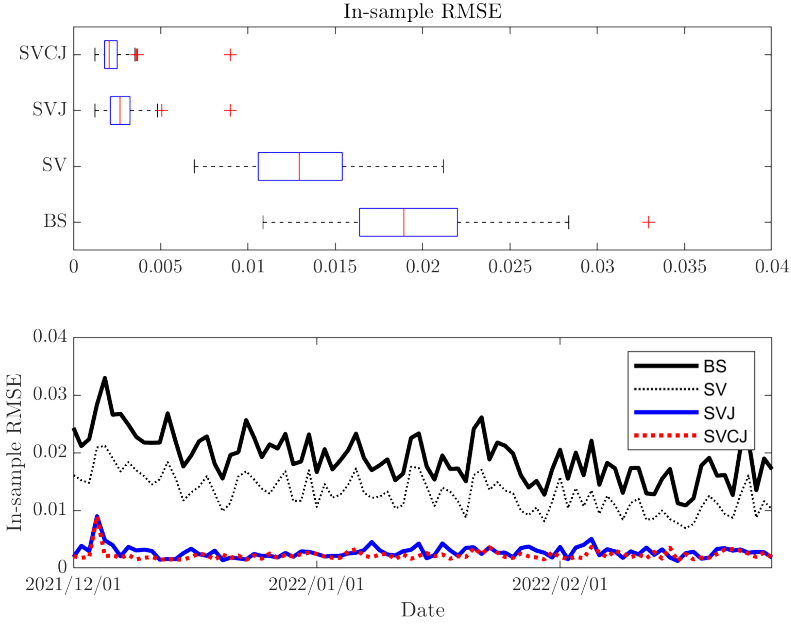


Fig. 5: **In-sample root-mean-squared errors.** The upper and lower panels display boxplots and time-series plots for in-sample RMSEs in calibrating intraday inverse option prices under the BS, SV, SVJ, and SVCJ models.

Deribit_inverse_BTC_options

To examine whether the SVCJ model is prone to overfitting, we calculate the out-of-sample RMSE at day t by substituting the implied parameter vector at day t with the implied parameter vector at day $(t-1)$. We present boxplots and time series plots for the out-of-sample RMSE in Figure 6. The SVCJ model continues to produce the lowest out-of-sample RMSE, followed by the SVJ, SV, and BS models. Therefore, we may conclude that the SVCJ model does not suffer from introducing an additional parameter.

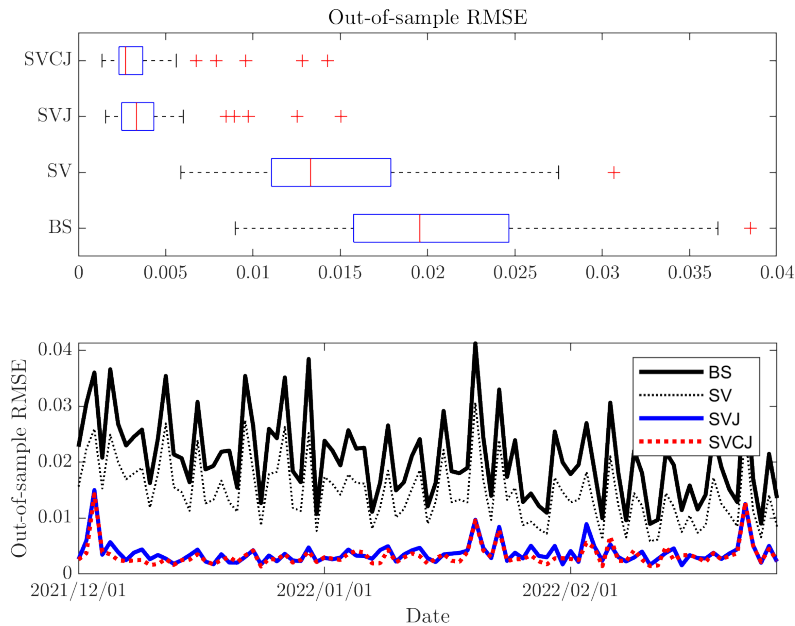


Fig. 6: **Out-of-sample root-mean-squared errors.** The upper and lower panels display boxplots and time-series plots for out-of-sample RMSEs in calibrating intraday inverse option prices under the BS, SV, SVJ, and SVCJ models.

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Implied parameters

Table 4 presents the average implied parameters for the BS, SV, SVJ, and SVCJ models using intraday BTC options calibrated on a daily basis. We also convert α and β to κ and θ for clearer presentations.

Table 4. Average implied parameters. This table presents the average implied parameters for the BS, SV, SVJ, and SVCJ models using intraday BTC options calibrated on a daily basis.

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	BS	SV	SVJ	SVCJ
σ	0.0278			
V_0		0.0013	0.0009	0.0009
κ		0.2203	1.5459	0.7425
θ		0.0006	0.0008	0.0008
σ_v		0.0216	0.0368	0.0489
ρ		0.1462	-0.3310	-0.6245
λ			0.0040	0.0033
μ_y			0.2359	0.3166
σ_y			0.1890	0.1267
ρ_j				-1.0821
μ_v				0.0206

Figures 7 to 15 provide time series plots of implied parameters for the BS, SV, SVJ, and SVCJ models. Generally, these implied parameters exhibit variability over time, suggesting that multiple parameter combinations may yield identical option prices. These observations echo the findings on the ill-posed problem for implied parameters in He et al. (2006). Indeed, the similarity in results offers a basis for further investigation on the stability of implied parameters and their impact on model performance.

In detail, Figure 7 displays the implied volatility from the BS model. For the BS model, the average implied volatility is 0.0278, which is approximately 53.11% ($= 100\sqrt{365} \times 0.0278\%$) on an annual basis. The average implied volatility from BTC options is significantly larger than the average implied volatility for options in traditional stocks. According to Goncalves and Guidolin (2006), in comparison to 13.2%, the average implied volatility (January 3, 1992, to June 28, 1996). This rather different corresponding values from Hang Seng Index options and Nikkei-225 Index options are 27.22% and 25.69% for the period from May 1998 to February 2005 (Taylor et al., 2010).

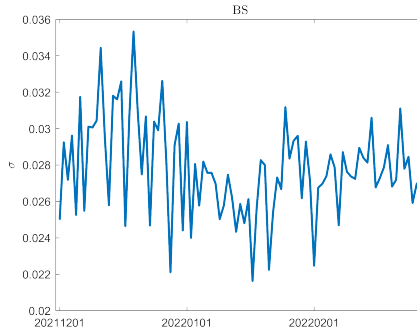


Fig. 7: **Implied parameters for the BS model.** This figure presents a time series plot of daily calibrated implied parameters for the BS model.  `Deribit_inverse_BTC_options`

Figure 8 compares the implied V_0 for the SV, SVJ, and SVCJ models. Since V_0 refers to the initial variance in the variance process, the SV model produces the highest average implied V_0 at 0.0013 (annually $68\% = \sqrt{0.0013 \times 365}$), higher than the SVJ and SVCJ models at 0.0009 (annually $57.3\% = \sqrt{0.0009 \times 365}$). SVJ and SVCJ models produce, on average, smaller V_0 because their jump components can compensate for the variation in the dynamics.

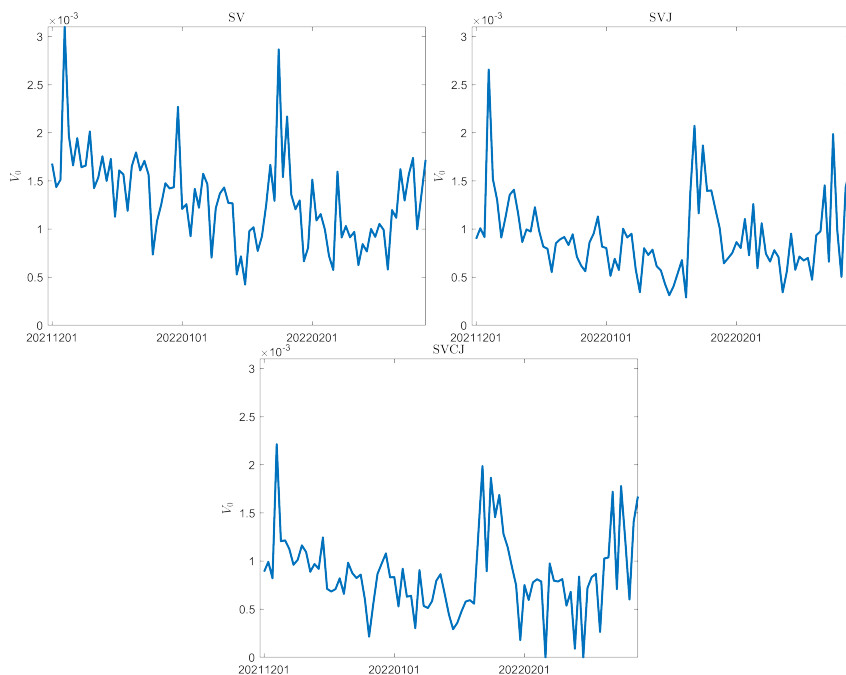


Fig. 8: Implied V_0 from the SV, SVJ, and SVCJ models

Figure 9 depicts the time series plots of the implied κ from the SV, SVJ, and SVCJ models. It is shown that, on average, the SVJ model produces the highest mean reversion parameter κ with an average value of 1.5459, followed by the SVCJ model with a value of 0.7425, and, the SV model with a value of 0.2203, see Table 4.

What do we learn from this plot? Small high, big high, medium high.

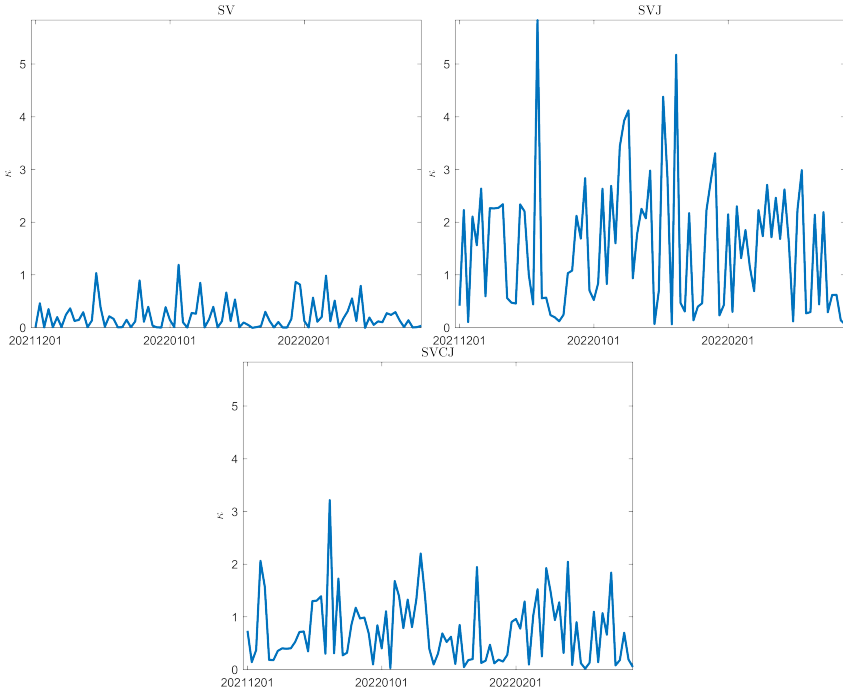


Fig. 9: Implied κ from the SV, SVJ, and SVCJ models

Figure 10 depicts the time series of implied θ for the SV, SVJ, and SVCJ models. The implied θ patterns for the SVJ and SVCJ models are similar, with an average value of 0.0008 (annually $54.03\% = \sqrt{0.0008 \times 365}$). In contrast, the SV model generates a lower average implied θ of 0.0006 (annually $46\% = \sqrt{0.0006 \times 365}$).

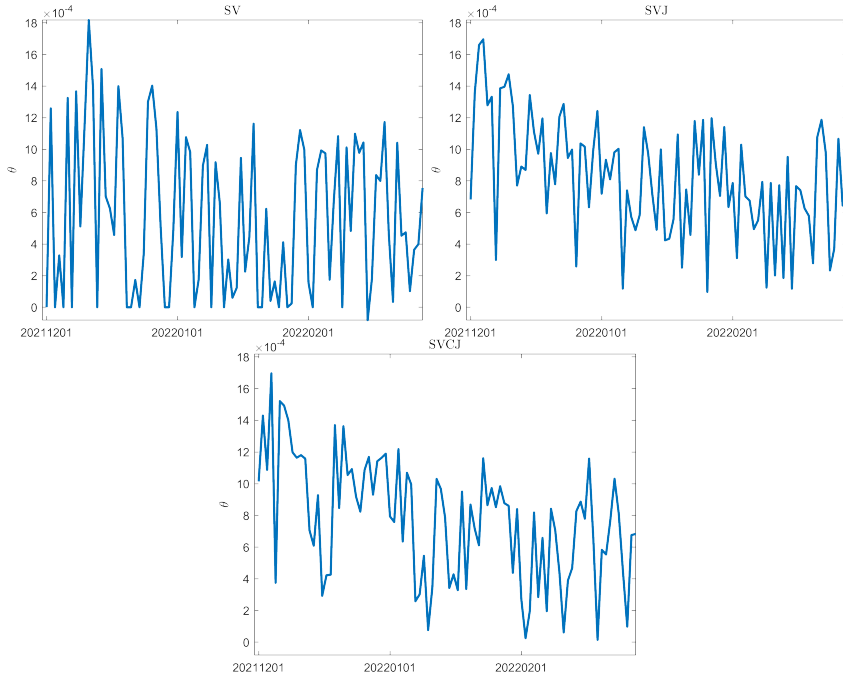


Fig. 10: Implied θ from the SV, SVJ, and SVCJ models

Figure 11 depicts time series plots of implied σ_v for the SV, SVJ, and SVCJ models. It appears that σ_v compromises with V_0 : while the SV model produces the highest average implied V_0 , it also has the lowest average value of σ_v at 0.0216. In contrast, the SVJ model has an average implied V_0 lower than the SV model and a higher average value of σ_v at 0.0368. Finally, the SVCJ model has an intermediate value of average implied V_0 and the highest average value of σ_v at 0.0489.

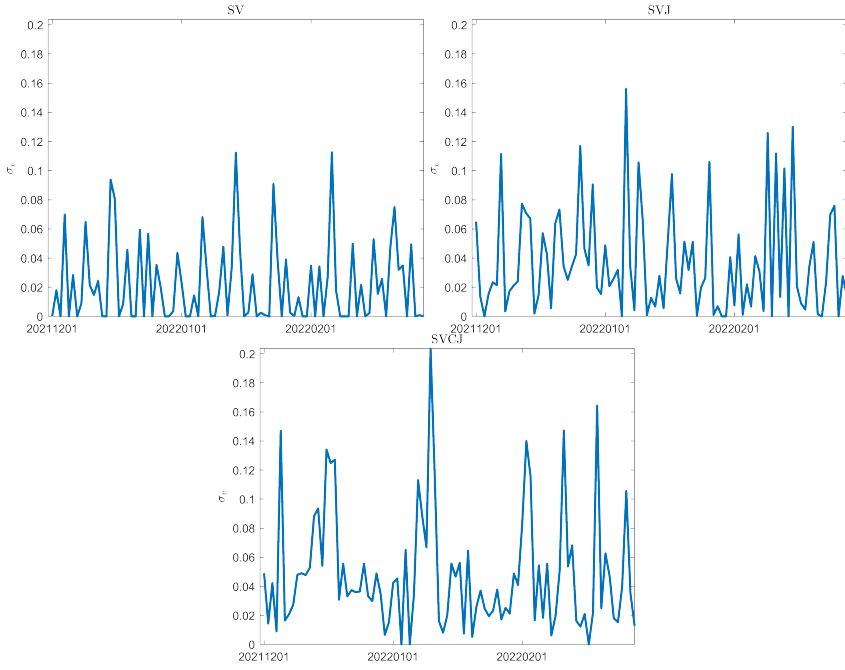


Fig. 11: Implied σ_v from the SV, SVJ, and SVCJ models

Interestingly, the implied ρ , representing the correlation between the two Wiener processes in the return and variance processes, differs substantially across the stochastic volatility models. On average, the SV, SVJ, and SVCJ models yield implied ρ of 0.1462, -0.3310 , and -0.6245 , respectively.

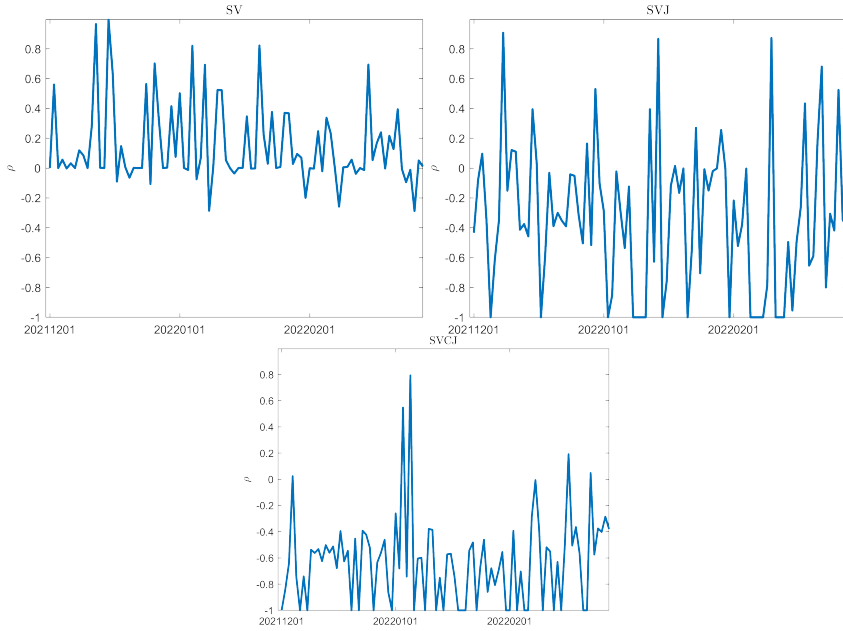


Fig. 12: Implied ρ from the SV, SVJ, and SVCJ models

We now analyze the implied parameters for the SVJ and SVCJ models. Since the implied jump intensity parameter (λ) remains constant over the study period for both models, we refrain from presenting their time series plots in the interest of brevity. The SVJ model generates an implied λ of 0.0040, corresponding to approximately 1.46 jumps per year ($1.46 = \sqrt{0.0040 \times 365}$). In comparison, the SVCJ model yields a marginally smaller implied λ value at 0.0033, translating to an average of 1.20 jumps annually (computed as $1.20 = \sqrt{0.0033 \times 365}$). This comparison highlights the differences in jump frequency between the two models, providing valuable insights for researchers and practitioners seeking to understand the behavior and characteristics of these models in the context of empirical finance applications.

We compare the implied values of μ_y and σ_y derived from the SVJ and SVCJ models, as illustrated in Figures 13 and 14, respectively. Our findings reveal a trade-off between the two models' implied values of μ_y and σ_y . Specifically, the SVJ model generates, on average, a lower implied value of μ_y at 0.2359 while exhibiting a higher value of σ_y at 0.1890. Conversely, the SVCJ model yields a higher average implied value of μ_y at 0.3166 but a lower value of σ_y at 0.1267.

This comparison offers valuable insights into the contrasting behavior of the two models to their risk and return characteristics, providing a foundation for further exploration into their suitability for a range of empirical finance applications.

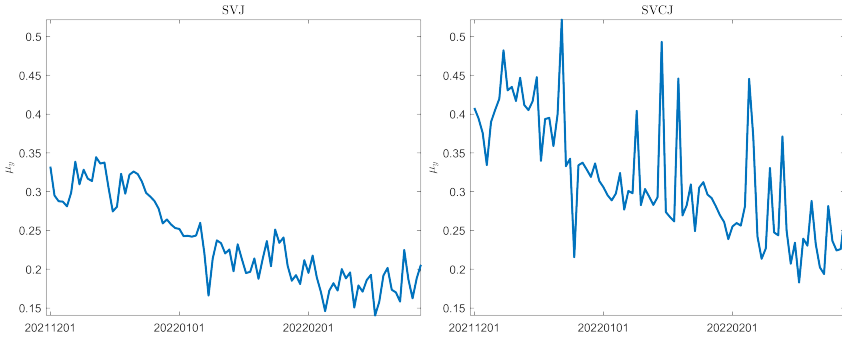


Fig. 13: Implied μ_y from the SVJ and SVCJ models

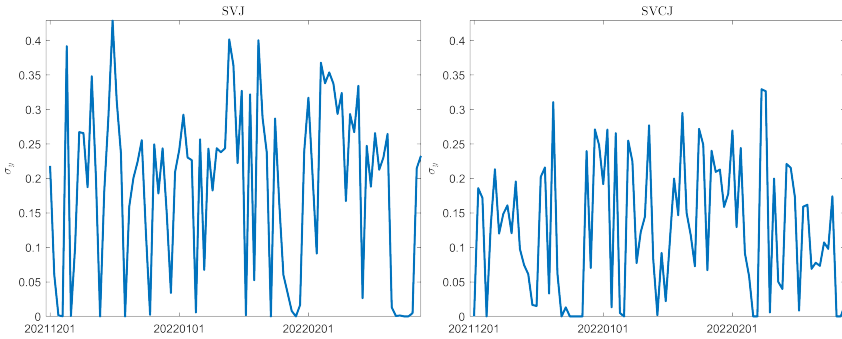


Fig. 14: Implied σ_y from the SVJ and SVCJ models

The SVCJ model's final analysis incorporates jump components in the variance process. We present the time series plots of implied ρ_j and μ_v in Figure 15, where most ρ_j values are negative and the implied values of μ_v exhibit variability over time.

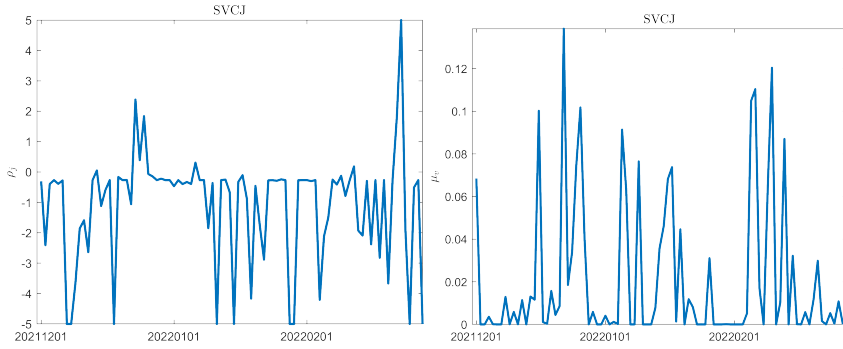


Fig. 15: Implied ρ_j and μ_v from the SVCJ model

Can we have a strong story here?

Dynamic delta hedging

In Section 4.1, we provide a general Delta formula that is applicable under complex dynamics. Equipped with this Delta, we implement dynamic delta hedging using inverse options and compare whether the SVCJ model outperforms its nested ones in Section 4.2.

The Delta formula

The Delta, denoted as Δ , of a financial option, represents how much the option price is affected by a change in the underlying asset's price. It is calculated as the ratio of the change in the option price to a slight change in the underlying asset price. With plain vanilla option prices, Matic et al. (2023) simulate complex dynamics for CCs to study hedging performance.

For the inverse option, the Delta is the price sensitivity to the initial underlying price. To emphasize the initial underlying price S_0 , we replace the parameter vector Θ by simply S_0 and denote $m(S_0; \omega, K, T)$ as the model price with option contract ω, K, T . Specifically,

$$\Delta = \frac{\partial}{\partial S_0} m(S_0; \omega, K, T).$$

In other words, when the underlying BTC index raises by one unit, the changes in the option price is Δ unit of BTC.

Calculating the Delta in a more complicated model is challenging with no closed-form option price formula. To tackle this difficulty, we apply the *parameter derivative* in Lyuu and Teng (2011) and Lyuu et al. (2019) to provide the Delta formula for the inverse option. The parameter derivative approach is generally applicable and simple to implement, and it allows us to provide a formula to calculate an unbiased Delta, as long as the stock prices at maturity can be generated as a function of S_0 and a random vector χ , i.e., $S_T = S_0 g(\chi)$. Let $I_A(x)$ denote the indicator function for set A that gives one if x belongs in A and gives zero otherwise. We rewrite the payoff

function of the inverse option as a product of a smooth function and an indicator function,

$$\begin{aligned}
 \wp(\omega, K, T) &= S_T^{-1} \max \{ \omega(S_T - K), 0 \} \\
 &= S_T^{-1} \omega(S_T - K) I_{\{ \omega(S_T - K) > 0 \}}(S_T) \\
 &= \omega \left(1 - \frac{1}{S_0} \frac{K}{g(\chi)} \right) I_{\{ \omega(S_T - K) > 0 \}}(S_T).
 \end{aligned}$$

With the parameter derivative, we can easily obtain the Delta formula:

$$\begin{aligned}
 \Delta &= e^{-rT} \mathbb{E} \left\{ \frac{\partial \wp(\omega, K, T)}{\partial S_0} \right\} \\
 &= e^{-rT} \mathbb{E} \left\{ \omega \left(\frac{1}{S_0^2} \frac{K}{g(\chi)} \right) I_{\{ \omega(S_T - K) > 0 \}}(S_T) \right\} \\
 &= e^{-rT} \mathbb{E} \left\{ \omega \left(\frac{1}{S_0} \frac{K}{S_T} \right) I_{\{ \omega(S_T - K) > 0 \}}(S_T) \right\}. \tag{14}
 \end{aligned}$$

Notably, the Delta formula's form is the same under the BS, SV, SVJ, and SVCJ models, but the Deltas differ from how S_T is generated. (14) can be simplified as

$$\Delta = \frac{\omega(e^{-rT}) K}{S_0} \mathbb{E} \left\{ \left(\frac{1}{S_T} \right) I_{\{ \omega(S_T - K) > 0 \}}(S_T) \right\}. \tag{15}$$

Because the expectation in the right-hand side of (15) is positive (assuming BTC would never has a negative price), the sign of inverse BTC call and put options depend on ω . Thus, inverse BTC call and put options have positive and negative Deltas.

Suppose S_T that varies from $0.1S_0$ to $10S_0$. The absolute value of Delta approximately falls within the range:

$$\frac{1}{10} \frac{e^{-rT} K}{S_0^2} \leq |\Delta| \leq 10 \frac{e^{-rT} K}{S_0^2}. \tag{16}$$

Hence, Δ is extremely close to zero in scale. Figure 16 depicts Deltas under the BS, SV, SVJ, and SVCJ models for different times to maturity calculated through (15) using simulation with a sample size of $N = 10,000$. Each panel presents Deltas with $T = 7$ (one week), $T = 28$ (four weeks), $T = 90$ (a quarter), and $T = 180$ (half year), respectively. We overlay a horizontal line in each panel to separate positive and negative Deltas for call and put options. We use implied parameters calibrated on January 1, 2022, as the parameter for each model. In addition, we set $S_0 = 47128.85$, which is the last price of BTC on January 1, 2022.

Deltas are minimal values at a magnitude of 10^{-5} , consistent with our observations in (16). Second, Deltas under the four models are incredibly close for short-term and mid-term inverse options, i.e., $T = 7$ and $T = 28$. Deltas under different models deviate for long-term options, i.e., $T = 90$ and $T = 180$. However, BS and SV models produce close Deltas, and the SVJ and SVCJ models produce almost identical Deltas.

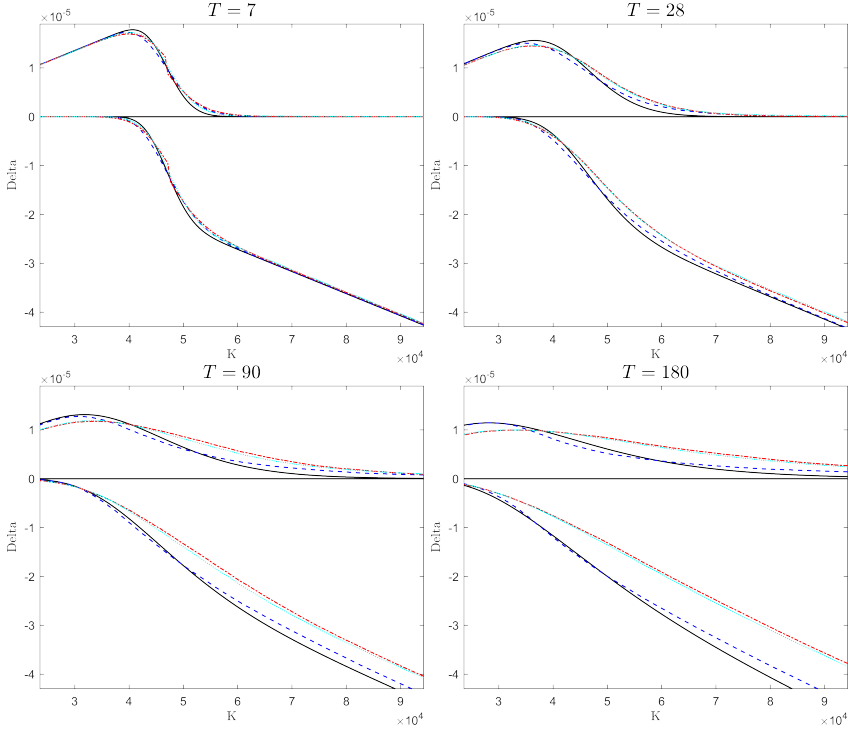


Fig. 16: **Deltas under the BS, SV, SVJ, and SVCJ models.** Each panel presents Deltas with different times to maturity: $T = 7$ (one week), $T = 28$ (four weeks), $T = 90$ (a quarter), and $T = 180$ (half year). We overlay a horizontal line in each panel. Lines above and below the horizontal line are Deltas for inverse BTC call and put options, respectively. The Deltas under the BS, SV, SVJ, and SVCJ models are in a black-solid line, an orange dashed line, a red dashed-dotted line, and a cyan dotted line, respectively.  `Deribit_inverse.BTC_options`

Hedging performance

To fully explore dynamic Delta hedging under various models with intraday transactions, we rearrange the data as follows. We pick up the last traded inverse options for each trading day, then, fixing a trading day in the study period and a maturity-strike combination, we check if this inverse option has consecutive trades to maturity. If so, we record this inverse option and implement dynamic delta hedging. Otherwise, we ignore it. We repeat the above procedure for each trading day during the study period and calculate the relative hedging errors at various maturity-money categories on each trading day in the study period.

We describe how to implement the dynamic Delta hedging for inverse options. Suppose we are hedging on a daily basis, $t = 1, \dots, T$. Given an option contract, let $P(t)$ denote its price denoted in BTC at time t . Let $\Delta(t)$ denote the Delta of

this option at time t . Consider a Delta-neutral portfolio that is composed of three ingredients:

1. A short position in an inverse option of value $P(t)$
2. A long position of $\Delta(t)$ number of BTC
3. An amount $B(t)$ in the money market denoted in USD

The Delta-neutral portfolio has value denoted in USD:

$$\Pi(1) = -P(1)S(1) + \Delta(1)S(1) + B(1), \quad (17)$$

where we set

$$B(1) = P(1)S(1) - \Delta(1)S(1)$$

to form a self-rebalancing portfolio with zero initial value. Then, the dynamic hedge procedures are repeated on a daily basis as follows. At day t , we adjust the position of BTC from $\Delta(t-1)$ to $\Delta(t)$ and obtain

$$\Pi(t) = -P(t)S(t) + \Delta(t)S(t) + B(t),$$

where the value changes in BTC are added to the money market account,

$$B(t) = e^{r(t-(t-1))} B(t-1) - S(t) \{ \Delta(t) - \Delta(t-1) \}.$$

Repeat the above procedures until the time to maturity T . The hedging portfolio value at maturity is $\Pi(T)$. To measure the performance of hedging, we define the relative hedging error, denoted by π , as

$$\pi = \frac{e^{-rT} \Pi(T)}{P(1)S(1)}.$$

Figure 17 summarizes the relative hedging error for inverse BTC call options with boxplots across nine maturity-moneyness categories. Because deep OTM inverse BTC call options do not have consecutive trades to maturity during our study period, we only show hedging results for short-term and ITM options. We find that distributions of relative hedging errors among the BS, SV, SVJ, and SVCJ models are indistinguishable. Moreover, the same phenomena for inverse BTC put options hold, as shown in Figure 18.

In summary, empirical results indicate that dynamic Delta hedging for inverse options with a more complicated model does not necessarily outperform simpler models, like eg., the BS. These results are consistent with existing literature for vanilla options Bakshi et al. (1997). The prediction error caused by a model is one reason hedging becomes indistinguishable across models. In the case of inverse options, as shown in Section 4.1, Deltas for short-term and mid-term options are incredibly close. On the other hand, Table 1 indicates that about 80%

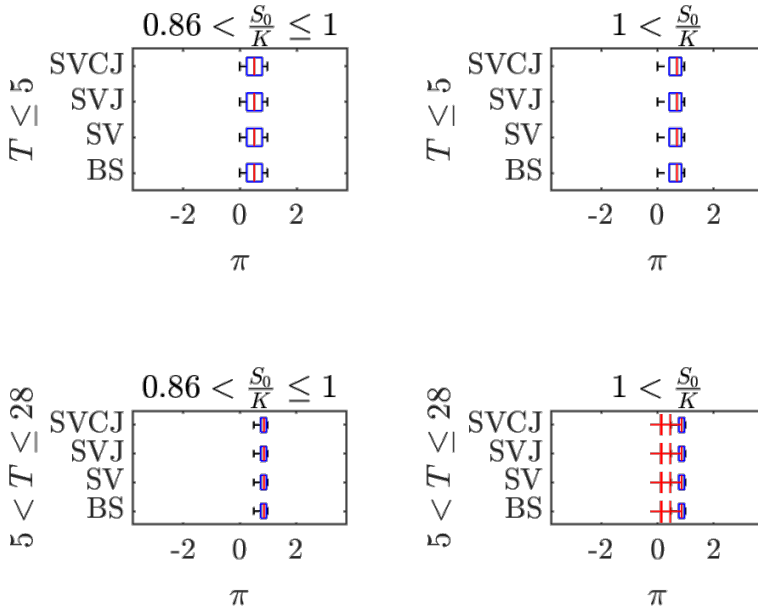


Fig. 17: **Relative hedging errors for inverse BTC call options.** This figure compares boxplots of relative hedging errors for inverse BTC put options at each maturity-moneyness category. [Deribit.inverse_BTC_options](#)

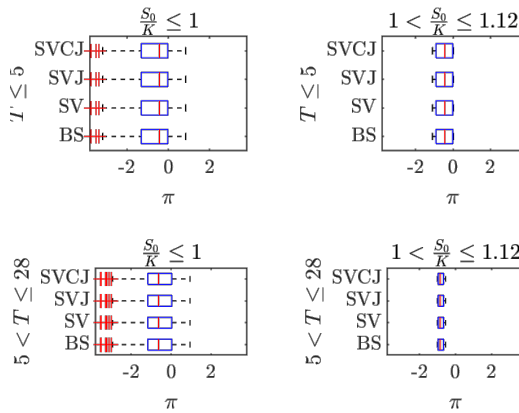


Fig. 18: **Relative hedging errors for inverse BTC put options.** This figure compares boxplots of relative hedging errors for inverse BTC put options at each maturity-moneyness category. [Deribit.inverse_BTC_options](#)

Conclusion

This paper investigates the intraday transactions of inverse BTC options traded on Deribit, focusing on the distinctive characteristics of these options. We propose a practical stochastic optimization approach to calibrate sophisticated models incorporating stochastic volatility and jumps. Our empirical analysis reveals that the SVCJ model provides a better fit than its nested models in both in-sample and out-of-sample pricing assessments. Moreover, we also provide the Delta hedging techniques enabling the implementation of dynamic hedging strategies. Notably, in the hedging exercise, the SVCJ model demonstrates comparable performance to simpler models when regarding relative hedging errors.

A comprehensive comprehension of cryptocurrencies and their derivatives is imperative, particularly in portfolio management, trading, and investor behavior. It is worth exploring opportunities for arbitrage arising from the slippery nature of implied volatility and investigating the put-call parity for inverse options. Furthermore, we acknowledge that multiple parameters can yield the same prices for inverse options, making it intriguing to examine the integration of historical data parameters during model calibration. Additionally, further research is needed to enable rapid and accurate pricing of inverse options and higher-order Greeks.

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Appendix

Table 5. Data descriptions. This table lists columns and their descriptions for the intraday transaction of Deribit inverse options through the Blockchain-Research-Center.com.

Index	Column	Type	Description
1	<i>q</i>	number	Quantity
2	<i>p</i>	number	Option price denoted in BTC
3	<i>s</i>	string	
4	<i>t</i>	integer	Timestamp (13 digit UNIX format, milliseconds since the UNIX epoch)
5	<i>d</i>	string	Date (YYYY/mm/dd)
6	<i>trade_seq</i>	integer	The sequence number of the trade within instrument
7	<i>trade_id</i>	string	Unique (per currency) trade identifier
8	<i>tick_direction</i>	integer	Direction of the "tick" (0=Plus Tick, 1 = Zero-Plus Tick, 2 =Minus Tick, 3 = Zero-Minus Tick)
9	<i>iv</i>	number	implied volatility
10	<i>instrument_name</i>	string	UNDERLYING-MATURITY-STRIKE(Call/Put)
11	<i>index_price</i>	number	the price of the underlying BTC denoted in USD
12	<i>direction</i>	string	